Homework 9

Please read Section 2.3 of Durrett. This section introduces the Poisson(λ) distribution and has many wonderful examples of probability calculations.

1. Our class has 15 students. I usually split the 15 of you into 5 groups of 3 students each for group work and quizzes. The order of the groups does not matter, since each group gets identical work. How many ways are there to split a class of 15 students into 5 groups of 3 students where the order of the groups does not matter? The number may be too large for your calculator to display exactly (without rounding). Write it in exact form (with factorials) and give whatever value your calculator provides.

This question is like questions from Homework 5 and (especially) Homework 7. Multinomial counting tells us that there are 15!/(3!3!3!3!3!) ways to divide the class of 15 people into 5 groups of 3 where the order of the groups matters. We would count this way if, for example, each group got a different task to work on in class or a different quiz. But the order of the groups does not matter, since each group gets the same in-class tasks and same quizzes. There are 5! ways to permute the 5 groups, and we divide by 5! to adjust for the fact that the order of the groups does not matter. This gives us

15!/(3!3!3!3!3!5!) = 1,401,400

ways to divide up the class. It's not too big for the calculator!

We can make the reasoning more precise by using the multiplication rule, as when we derived the formula for $C_{n, k}$ on the first day of class and as in my solutions for Homework 7. Let N be the number of ways to divide the class into 5 groups of 3 where the order does not matter. Imagine doing the multinomial counting above (5 groups of 3 where the order matters) in two stages: first divide the class into 5 groups of 3 where the order of the groups does no matter, and then choose a permutation of those 5 groups to put them in order. According to the multiplication rule, there are N(5!) ways to do that. Therefore, we have N(5!) = 15!/(3!3!3!3!3!), and so N = 15!/(3!3!3!3!3!5!).

2. Please work Exercise 44 from **Chapter 1**. This exercise is about the casino game roulette.

For these calculations, our outcomes are the 38 slots on the Roulette wheel (1–36, 0, and 00). We treat each outcome as equally likely. The probability of an event (set of outcomes) is the number of outcomes in the even divided by 38, the total number of equally likely outcomes.

a. Let X be the random variable taking the value 1 for the outcomes 1–18 and –1 for the other outcomes. We thus have P(X = 1) = 18/38 and P(X = –1) = 20/38. Therefore, E[X] = (1)(18/38) +

(–1)(20/38) = –2/38. (The units of measurement are dollars.)

b. Let Y be the random variable taking the value 2 for the outcomes 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36 (multiples of 3) and –1 for all other outcomes. We thus have P(X = 2) = 12/38 and P(X = –1) = 26/38. Therefore, E[X] = (2)(12/38) + (–1)(26/38) = 24/38 – 26/38 = –2/38.

c. Let Z be the random variable taking the value 35 for the outcome 7 and –1 otherwise. We thus have P(X = 35) = 1/38 and P(X = –1) = 37/38. Therefore, E[X] = (35)(1/38) + (–1)(37/38) = 35/38 – 37/38 = –2/38.

All the bets have the same expected value!

3. Please work Exercise 38 from Chapter 2. To solve this problem, answer the following questions:

a. Imagine the chance experiment of rolling two dice 24 times. Let N be the random variable counting the number of double 6's in your 24 trials. Match N with the binomial-distribution story. What is the distribution of N?

Each trial (rolling two dice) has a 1/36 chance of "success" (rolling double 6's). We treat the trials as independent. The total number N of successes (double 6's) in 24 independent trials has the binomial(24,1/36) distribution. In symbols: N ~ binomial(24,1/36).

b. What is the probability of the event (N ≥ 1) according to the formula for binomial probabilities?

The complement of the event (N ≥ 1) is the event (N = 0). We thus have P(N ≥ 1) = 1 – P(N = 0). According to the formula for binomial probabilities, P(N = 0) = 1 – $(1 – 1/36)^{24}$ = 1 – $(35/36)^{24}$ ≈ 0.4914. You calculated this probability on Homework 6 (before you learned to use the binomial distribution).

c. The distribution of N is approximately a Poisson(λ) distribution. What is λ?

For large n and small p, the binomial(n, p) distribution approximately matches the Poisson(λ) distribution. In this case, we have λ = (24)(1/36) = 2/3.

d. Use this Poisson approximation to the distribution of N to approximate P(N ≥ 1). What is your approximate value? How close is it to the value you found using the binomial formula?

Using the same reasoning as above, we have P(N ≥ 1) = 1 – P(N = 0) ≈ 1 – $e^{-2/3}$ ≈ 0.4866. This approximation is pretty close to the binomial probability, even though n is not very large and p is not very small.

4. Please work Exercise 39 from Chapter 2. The answer in the book is 0.3297. Follow the same procedure as for Exercise 38: What random variable will help you answer this question? This

random variable has a binomial distribution. What are n and p? The distribution is approximately a Poisson(λ) distribution. What is λ? Use the Poisson approximation to estimate the probability of getting at least one three of a kind if you play 20 hands of poker.

Let N be the random variable that counts the number of three-of-a-kinds we get in 20 hands of poker. According to what Durrett tells us, the probability of a three-of-a-kind in a single trial (one hand) is about 1/50. Assuming that our poker hands are independent, the number N of three-of-a-kind hands in 20 hands has (roughly) the binomial(20, 1/50) distribution. That distribution is approximately Poisson(λ) distributed with $\lambda = (20)(1/50) = 0.4$.

We have $P(N \geq 1) = 1 - P(N = 0)$ and $P(N = 0) \approx e^{-0.4}$. Therefore, $P(N \geq 1) \approx 1 - e^{-0.4} \approx 0.3297$.

A bit of research online suggests that 20 hands is about what one would play in an hour.

5. Please work Exercise 43 from Chapter 2. The textbook answer is 0.1889. Follow the same procedure as from Exercises 38 and 39.

There are 210,850,582 Powerball lottery tickets sold. Each ticket has a 1/80,000,000 chance of winning. The number N of winning tickets has the binomial(210,850,582, 1/80,000,000) distribution. This distribution is approximately Poisson(λ) for $\lambda = (210{,}850{,}582)(1/80{,}000{,}000) \approx 2.6356$. We thus have

$P(N = 1) \approx \lambda^1 /1!\, e^{-\lambda} \approx 0.1889$.

6. The following question will get us ready for Exercise 51 in Chapter 2, which you will complete in the next homework assignment. In that question, Durrett tells us that "[b]ooks from a certain publisher contain an average of 1 misprint per page." Let N be the number of misprints on a randomly chosen page of a book from this publisher. We will model the number of misprints on the page as a Poisson(λ) random variable.

a. What value should we use for λ?

We will use $\lambda = 1$ matching the rate (1 error per page on average) from the question. Here is a justification for that choice: as you read in Section 2.3, if N is a Poisson(λ) random variable, then $E[N] = \lambda$. We think of E[N] informally as the long-run average of independent data drawn from this random variable. That long-run average should match the given average number of errors per page.

This method of matching parameters of a distribution to data is called "the method of moments." For more complicated distributions, we might match $E[N]$, $E[N^2]$, $E[N^3]$, and more to account for all the parameters of the model. The method of moments works well in simple cases (like this one). Statisticians fitting probability models to data use more sophisticated methods (such as "maximum likelihood") in more complex situations.

b. What is the probability that a randomly chosen page has at least five misprints? Calculate a decimal value.

We want to calculate P(N ≥ 5). The complement of the event (N ≥ 5) is (N < 5), which is the same as the event (N ≥ 4), since the possible values of N are 0, 1, 2, 3, 4, 5, … . The event P(N ≤ 4) is the disjoint union of the events (N = 0), (N = 1), (N = 2), (N = 3), and (N = 4), and so

$$P(N \leq 4) = P(N=0) + P(N=1) + P(N=2) + P(N=3) + P(N=4) \text{¿} e^{-1} + \frac{1^1}{1!}e^{-1} + \frac{1^2}{2!}e^{-1} + \frac{1^3}{3!}e^{-1} + \frac{1^4}{4!}e^{-1}$$

$$\text{¿} \frac{65}{24}e^{-1} \approx 0.99634.$$

We thus have $P(N \geq 5) = 1 - \frac{65}{24}e^{-1} \approx 0.003660$.